

Smart Traffic Signal Controller

Roll NO. 2k23/ CSM /135 (Hayat Shah)

Task 1: PEAS Specification

- Performance Measures

Average vehicle waiting time	Lower is better
Intersection throughput (vehicles/hour)	More vehicles passing per hour is better
Average queue length	Shorter queues are better
Traffic violation detection accuracy	More accurate is better
Emergency vehicle priority success rate	How reliably ambulances/fire trucks get priority
False violation alert rate	Lower is better

-Environment

Urban road intersections containing:

Cars, motorcycles, buses, trucks

Pedestrians, cyclists

Emergency vehicles

Traffic police authorities

Varying weather conditions

Day and night traffic

- Actuators

1. Traffic light controller
2. Pedestrian crossing signals
3. Network communication system to coordinate nearby intersections
4. Police notification system
5. Violation report generation system

- Sensors

1. High-resolution RGB cameras
2. Automatic Number Plate Recognition (ANPR) cameras
3. Inductive loop vehicle detectors embedded in roads
4. Radar/LiDAR traffic sensors
5. Emergency vehicle transponder receivers

6. Weather sensors

7. GPS data from connected vehicles or buses

- Task 2: Environment Classification

Partially Observable

The AI cannot observe every vehicle and future traffic event because some information may be hidden by obstacles, weather, or sensor limitations.

Multi-Agent

Multiple intersections and many road users interact with each other.

Stochastic

Driver behavior, accidents, weather changes, and traffic violations introduce uncertainty.

Sequential

Each traffic-light decision affects future traffic flow and congestion levels.

Dynamic

Traffic conditions continuously change while the AI is operating.

Continuous

Vehicle speeds, traffic density, and signal timings change continuously over time.

- Task 3: Critical Analysis

- Question 1

Greatest Challenge: Stochastic Environment

The stochastic nature of the environment presents the greatest challenge. Traffic conditions can change unexpectedly due to accidents, weather conditions, road construction, and human behavior. In addition, traffic violations such as red-light jumping can occur at any time, making it difficult for the AI system to predict future states accurately. Therefore, the AI must continuously adapt its decisions using real-time data.

- Question 2

Utility Function = $0.5 \times \text{TrafficFlow} + 0.3 \times \text{SafetyScore} + 0.2 \times \text{ViolationDetectionAccuracy}$

Where:

TrafficFlow = number of vehicles moving efficiently

SafetyScore = safety level at intersections

ViolationDetectionAccuracy = accuracy of identifying traffic offenders

Effect of Doubling the Safety Weight

If the SafetyScore weight increases from 0.3 to 0.6, the AI will prioritize safety over traffic speed. It may allow longer red-light intervals, increase pedestrian crossing times, and reduce the risk of accidents even if traffic flow becomes slightly slower.

STRUCTURED REPRESENTATION: -----

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